

Department of Computer Sciences

CS-1005- Discrete Structures

Semester: Fall 2024

Assignment #2

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Due Date: As mentioned on Google Classroom

Max Score 144

Note:

1. Please submit handwritten solutions to all of the questions given below.
2. Scan your handwritten solution and upload to Google class room.
3. No extension in deadline.
4. Plagiarism would result in zero marks.

Question 1: Sequence and Summations (1 hour)

- a. Consider the sequence defined by $a_n = 3n + 2$. [20 marks]
 - i. Find the first five terms of the sequence.
 - ii. Determine the general/explicit formula for the sum of the first n terms of this sequence.
- b. Provide explicit formula for the n th term for the sum of first n terms of the sequence 0,2,4,6, 8,... $2n$
- c. Evaluate the following sums and products:[10 marks]
 - i. $\sum_{k=1}^{10} (2k + 1)$
 - ii. $\prod_{k=1}^5 (k^2 + 1)$
- d. Simplify the following expressions:[10 marks]
 - i. $10! / (8! * 2!)$
 - ii. $\sum_{k=0}^5 1/k!$
- e. provide a non-recursive formula that generates the terms of an integer sequence that begins with the given list 3,6,12,24,48,96,192,... . Also provide next 3 terms using the formula.

Question 2: Induction (1 hour)

- a. Use the formula for the sum of the first n integers and/or the formula for the sum of a geometric sequence to evaluate the sums [15 marks]
 - i. $5+10+15+20+\dots+300$
 - ii. $3+3^2+3^3+3^4+\dots+3^n$
 - iii. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$
- b. You have two parents, four grandparents, eight great-grandparents, and so forth. If all your ancestors were distinct, what would be the total number of your ancestors for the past 40 generations (counting your parents' generation as number one)?
(Hint: Use the formula for the sum of a geometric sequence. [10 marks])
- c. Prove that 3 divides $n^3 + 2n$ whenever n is a positive integer. [5 marks]

Question 3: Correctness of Algorithms

The following **while** loop implements a way to multiply two numbers that was developed by the ancient Egyptians.

[Pre-condition: A and B are positive integers, $x = A$, $y = B$, and $product = 0$.]

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while ( $y \neq 0$ )
   $r := y \bmod 2$ 
  if  $r = 0$ 
    then do  $x := 2x$ 
            $y := y \div 2$ 
    end do

  if  $r = 1$ 
    then do  $product := product + x$ 
            $y := y - 1$ 
    end do
end while

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[Post-condition: $product = AB$]

1. Make a trace table to show that the algorithm gives the correct answer for multiplying [10 marks]
 $A = 14$ times $B = 19$.
2. Prove the correctness of this loop with respect to its pre- and post-conditions by using the loop invariant [5 marks]
 $I(n)$: " $xy + product = AB$."

Question 4: Recursive Sequence

- a. Find the first four terms of each of the recursively defined sequences [5 marks]
 $V_k = V_{k-1} + V_{k-1} + 1$, $V_1 = 1$, $V_2 = 3$
- b. **Tower of Hanoi:** Suppose that in addition to the requirement that they never move a larger disk on top of a smaller one, the priests who move the disks of the Tower of Hanoi are also allowed only to move disks one by one from one pole to an adjacent pole. Assume poles A and C are at the two ends of the row and pole B is in the middle. Let a_n = the minimum number of moves needed to transfer a tower of n disks from pole A to pole C [10 marks]
 - i. Find a_1, a_2, a_3, a_4
 - ii. Find the recurrence relation for a_1, a_2, a_3, a_4
- c. Considering recurrence relation of Fibonacci number $F_0 = 1$, $F_1 = 1$, and prove the following [10 marks]
 $F_0^2 + F_1^2 + F_2^2 + \dots + F_n^2 = F_n \times F_{n+1}$
- d. Suppose the population of a country increases at a steady rate of 3% per year. If the population is 50 million at a certain time, what will it be 25 years later? Provide a non-recursive formula [10 marks]
- e. As shown in Example 5.6.8, if a bank pays interest at a rate of i compounded m times a year, then the amount of money P_k at the end of k time periods (where one time period = $1/m$ th of a year) satisfies the recurrence relation $P_k = [1 + \frac{i}{m}] P_{k-1}$ with

initial condition P_0 = the initial amount deposited. Find an explicit formula for P_n . [10 marks]

Question 5 Counting

- a. How many bit strings of length 10 contain [6marks]
 - i. exactly four 1s?
 - ii. at most four 1s?
 - iii. at least four 1s?
- b. A professor writes 40 discrete mathematics true/false questions. Of the statements in these questions, 17 are true. If the questions can be positioned in in any order, how many different answer keys are possible? [4 marks]
- c. How many license plates can be made following the scheme of Islamabad excise and taxation [4 marks]
 - i. consisting of three letters followed by three digits contain no letter or digit twice?
 - ii. consisting of three letters followed by three digits contain and letters and digits may be repeated